

Universität Bern / Zürcher Hochschule der Künste

CAS in AI for Creative Practice

2025 / 2026

Module 4

AI for Imaging

DRAFT

Chat-room.ch 2.0

Paper by

Anet Nyffeler

14-656-029

Handed in

Dr. Daniel Bisig, Dr. Christopher Salter

Zürich, xx.2026

Contents

1	ABSTRACT	4
2	THEORETICAL BACKGROUND	4
2.1	Translation science and assistive communication technologies	4
2.1.1	Intersemiotic and multimodal translation	4
2.1.1.1	Jakobson's intersemiotic translation	5
2.1.1.2	Intermodal / multimodal translation AVT (audiovisual translation research)	6
2.1.2	Functionalist (Skopos) perspectives and norms	6
2.1.2.1	Skopos theory / functionalism	7
2.1.2.2	Norms and ethics in translation	8
2.1.3	Relevance, effort, and cognitive load	8
2.1.3.1	Relevance theory & information management	9
2.1.4	Evolution of Assistive Communication Technologies	10
2.1.5	Role of AI in modern speech recognition evolution	11
2.1.5.1	End-to-end acoustic modeling	11
2.1.5.2	Joint acoustic–language modeling	11
2.1.5.3	Massive data and self-/semi-supervised learning	11
2.1.5.4	On-device and edge deployment	11
2.1.5.5	Multilingual and code-switching capabilities	12
2.1.5.6	Integration into accessibility ecosystems AI-driven ASR is now:	12
2.1.6	AI-based speech recognition and its limitations for inclusive design	12
2.2	Evolution and semantic shift of “chatroom”	14
2.2.1	Etymology and Origin	14
2.2.2	Early Definitions and Canonical Meaning	15
2.2.3	Cultural and Social Function (1990s–early 2000s)	15
		2

2.2.3.1	A Social Experiment	15
2.2.3.2	A Democratic Public Space	15
2.2.3.3	A Site of Anxiety	15
2.2.4	Semantic Shift and Decline	16
2.2.5	Contemporary Usage and Reframing	16
2.2.6	Meaning Change in Summary	17
2.2.7	Conceptual Insight	17
2.3	Reclaiming the “chatroom” in the age of multimodal, barrier-aware AI	18
2.4	Cybernetic art, sensing machines, and <u>chat-room.ch 2.0</u>	19
2.4.1	Cybernetic and Biofeedback Lineages in the Design of “chat-room.ch 2.0”	21
2.5	Language, space, and interactive multimodal platforms in <i>chat-room.ch 2.0</i>	22
2.6	Key findings of article from Jucker et al. (2018) on IMPs for <i>chat-room.ch 2.0</i>	24
2.6.1	Core Framework: Degrees of Spatial Structuring	24
2.6.2	IMP-Specific Insights: Layered Spatial Complexity	24
2.6.3	Actionable Ties to the Design of chat-room.ch 2.0	25
2.6.4	Theoretical Positioning (2026 Update)	26
2.7	Language, Space, and Interactive Multimodal Platforms in <i>chat-room.ch 2.0</i>	26
3	THE AUTHOR	27
4	PRACTICAL DESIGN OF CHAT-ROOM 2.0	28
4.1	Sensors	28
4.2	Access and inclusion	28

1 Abstract

chat-room.ch 2.0 is a multimodal chatroom in which an AI moderator and interpreter enables people, with or without impairments, to participate in a shared conversational space using the input modalities they choose (speech, text, gesture, or sensor-based signals). The piece explores the role of the AI moderator and interpreter and how it differs from the role AI has in applications such as WhatsApp and WeChat.

chat-room.ch 2.0 tests "doing space" on a web-based IMP, reclaiming the chatroom as moderately structured, sensor-rich architecture where diverse bodies co-produce conversational space.

"I'm taking the methods from barrier-free translation practice (STTI, respeaking, audio description, accessible web design) and re-situating them in a creative, sensor-driven conversational environment, where smartphones serve as inclusive sensing devices rather than just screens." Anet Nyffeler

2 Theoretical background

2.1 Translation science and assistive communication technologies

Barrier-free communication is fundamentally a problem of **translation**: not only between languages, but between different sensory channels and representational systems. For this project, it is useful to situate this work within three strands of translation theory:

intersemiotic translation, multimodal translation, and functionalist (Skopos) approaches.

Visitors witness **real-time translation**

- translating **embodied signals** (voice, breath, motion, possibly sign/gesture) into **audible and visual patterns** in a shared space.
- This is intersemiotic translation grounded in barrier-free communication research (speech-to-text, audio description, respeaking, STTI, etc.).

Therefore, the author investigates in translation theories applied to accessibility tech.

2.1.1 Intersemiotic and multimodal translation

Roman Jakobson distinguished three types of translation: **intra lingual** (within one language), **inter lingual** (between languages), and **inter semiotic**, which he defined as trans-

lation “from one sign system into another.” Many accessibility technologies fall squarely into this last category. Audio description translates visual information into spoken narration; subtitles for the deaf and hard of hearing translate speech and sound effects into written text and symbols; sign language interpreting for broadcast media translates between a visual-gestural system and oral/written language. Braille technologies translate between tactile code and print or speech.

From this perspective, barrier-free communication is not an afterthought but a central case of intersemiotic translation. For a system like *chat-room.ch* 2.0, every mapping—breath or motion into sonic patterns, speech into text, text into haptic feedback—can be framed as an intersemiotic translation pipeline. These pipelines are designed choices about what aspects of embodied signals become communicatively salient in the shared space.

This aligns with more recent work on **multimodal translation** in audiovisual translation and accessibility. Media texts are inherently multimodal, combining image, language, sound, and movement. Accessibility interventions are therefore multimodal remixes: they select, condense, and re-present parts of the multimodal source so that people with different sensory profiles can still construct a coherent mental model. This is evident in the way audio describers decide which visual elements to verbalize, or how subtitlers decide which sounds and prosodic nuances to represent in text.

In *chat-room.ch* 2.0, the “turns” in conversation are no longer only verbal; they are multimodal contributions: a hum translated into a melodic figure, a sequence of device tilts translated into rhythm, a spoken sentence translated into text and sound. The system’s job is not to mirror raw sensor data, but to choose multimodal forms that support participation and recognizability for others.

2.1.1.1 Jakobson’s intersemiotic translation

- Roman Jakobson distinguished:
 - Intralingual translation (within one language),
 - Interlingual (between languages),
 - Intersemiotic (between sign systems).
- Accessibility tech lives largely in the intersemiotic space:
 - Audio description: visual → spoken verbal.

- SDH/subtitles for the deaf and hard of hearing: speech + soundscape → written text + symbols.
- Sign ↔ text / speech: gestural-visual language ↔ spoken/written language.
- Braille-to-speech: tactile code → speech.

For chat-room.ch 2.0:

Every pathway like “breath → spectral line”, “gesture → sound pattern” is an intersemiotic translation. This gives a clean theoretical label: the author designing *intersemiotic translation pipelines* for conversation.

<https://journals.openedition.org/signata/1223>

2.1.1.2 Intermodal / multimodal translation AVT (audiovisual translation research)

AVT and accessibility research extend this to multimodal translation:

- Media combine modes (image, sound, text, movement). Accessibility adaptations are multimodal remixes:
 - Audio description: chooses which visual cues to verbalize, how detailed, in which register.
 - SDH: chooses which non-speech audio to represent and how (music tags, speaker IDs, sound effects).
 - Live respeaking/STTI: spoken input → edited written output in near real-time, balancing accuracy, condensation, and readability.

For chat-room.ch 2.0:

The author explicitly borrows AVT concepts: condensation, selection, salience. The designed system doesn't pass every raw sensor sample; it condenses them into meaningful conversational cues (e.g. sustained calm breathing → “soft drones”, sudden movement → percussive bursts).

<https://dialnet.unirioja.es/servlet/articulo?codigo=10277620>

2.1.2 Functionalist (Skopos) perspectives and norms

Skopos theory suggests that translation should be evaluated primarily by how well it achieves its purpose in the target context. In accessibility, this has shifted thinking away from literal rendering towards functional adequacy. Audio description might prioritize narrative coherence or emotional tone rather than exhaustively cataloguing visual details.

Live speech-to-text interpreting prioritizes readability and latency over exact reproduction. Simplified or “easy-to-read” texts trade lexical richness for comprehensibility.

Applying this to accessibility technologies means that each intersemiotic mapping needs an explicit **purpose**. In a conversational environment, the purpose of translating a user’s sensor signals into sound is not to provide medically precise physiological data but to offer a legible, expressive presence to other participants. Similarly, the purpose of text-to-speech output is to ensure that visually impaired users can follow the conversational structure, not to emulate every typography nuance. Skopos provides a principled language to justify simplifications and stylizations: choices are guided by communicative function and user needs rather than by fidelity to a “source modality.”

Translation norms and ethics also play a critical role. Accessibility work is framed by norms against patronizing language, against erasing difference, and for transparency about limitations. This becomes highly pertinent as soon as translation is performed by technical systems. An AI “moderator” that nudges under-participating tables or gently interrupts monopolizing voices can be seen as a functional analogue of a human facilitator; however, its interventions must be designed in line with norms of respect, non-coercion, and user agency.

2.1.2.1 Skopos theory / functionalism

- Skopos theory: translations should be judged by their purpose (skopos) in the target context, not by word-for-word equivalence.
- In accessibility:
 - Audio description’s purpose might be to enable narrative comprehension, or to convey emotional tone, not to mirror every visual detail.
 - SDH might aim for readability under time constraints, not full verbatim transcription.
 - Sign ↔ text systems often prioritize intelligibility and speed over full linguistic subtlety.

For [chat-room.ch](#) 2.0:

The author defines clear Skopoi for each mapping:

- For a non-speaking participant: sensor-to-sound translation’s purpose is participation and recognizability, not faithful physiological measurement.

- For the AI moderator: text-to-intervention mapping’s purpose is keeping conversation conditions healthy, not maximizing information throughput.

That gives the author a principled way to argue why to simplify or stylize signals.

2.1.2.2 Norms and ethics in translation

- Accessibility translation has strong ethical dimensions:
 - Avoid patronizing or infantilizing language.
 - Respect users’ autonomy and privacy.
 - Be transparent about limitations and “blind spots” of the adaptation.
- In tech, this maps to:
 - Respectful, non-diagnostic labeling (e.g. not treating stress patterns as “pathology”).
 - Informed consent on sensor use.
 - Not exposing participants’ data or states without control.

For chat-room.ch 2.0:

The author explicitly positions the AI as a non-dominating mediator, consistent with ethical norms: it nudges, prompts, redistributes turns, but doesn’t classify or judge users’ states.

2.1.3 Relevance, effort, and cognitive load

Pragmatic approaches such as relevance theory bring in the dimension of **cognitive effort**. Accessible translations strive for representations that provide meaningful contextual effects without overloading the user’s processing capacity. This is explicit in guidelines for subtitles (constraints on reading speed, line length) and in plain language and “easy language” recommendations (short sentences, clear information structure).

For chat-room.ch 2.0, this suggests that sensor-to-sound mappings and visualizations should be low in interpretive overhead. A simple, consistent sonic motif tied to each participant is more effective than a complex, constantly shifting mapping. A minimal visual representation of tables and activity is easier to parse—and easier to render in screen readers and Braille displays—than a complicated interface.

Taken together, these theoretical frames support a view of the piece as **deliberate multimodal translation design**: sensors and AI are not opaque “black boxes,” but tools for

constructing intersemiotic mappings with clear communicative purposes and ethical constraints. This gives the author a solid bridge from translation science to accessibility technology and to the specific design decisions in the piece.

2.1.3.1 Relevance theory & information management

- Relevance theory: we aim for optimal balance between cognitive effort and contextual effect.
- Accessibility:
 - SDH/adapted texts strive for formulations that give the most “bang for buck” for readers with limited processing resources.
 - Audio description chooses details that best help build a coherent mental model.

For chat-room.ch 2.0:

The sensor translations in the piece should be low-cognitive-load and high-effect:

- A simple recurring sonic motif for a user is easier to track than complex mappings.
- Minimal visual UI (just table positions and spectral lines) is easier for all, especially those using screen readers or with cognitive load limitations.

2.1.4 Evolution of Assistive Communication Technologies



This slide makes clear that [chat-room.ch](#) 2.0 builds on decades of intersemiotic, assistive translation technologies.

2.1.5 Role of AI in modern speech recognition evolution

Modern speech recognition has been transformed by AI, especially by deep learning. Historically, ASR relied on **hand-crafted statistical models** (e.g. Hidden Markov Models plus Gaussian mixture models) and carefully engineered features (MFCCs, language models). Accuracy plateaued for years, especially in noisy environments and for spontaneous speech.

Deep learning altered this in several ways:

2.1.5.1 End-to-end acoustic modeling

Neural networks (first deep feedforward nets, then recurrent and convolutional architectures, now transformers) learned to map acoustic features directly to phonemes, characters, or wordpieces. Instead of manually designed Gaussian mixtures, a single model can learn complex temporal and spectral patterns from large amounts of data. This significantly improved robustness and accuracy, especially for conversational speech and under variable conditions.

2.1.5.2 Joint acoustic–language modeling

Large neural models can jointly encode acoustic and linguistic context, making better use of long-range dependencies. Sequence-to-sequence models with attention, and later transformer-based models, integrate what used to be separate components (acoustic model, pronunciation model, language model) into more unified architectures. This enables more flexible adaptation to different domains and speaking styles.

2.1.5.3 Massive data and self-/semi-supervised learning

AI methods can exploit huge amounts of unlabeled or weakly labeled audio. Pretraining models on large speech corpora and then fine-tuning for specific tasks reduces the need for hand-transcribed data. This is crucial for low-resource languages and for specialized domains (e.g. conversational speech in noisy environments, accented speech, or clinical contexts).

2.1.5.4 On-device and edge deployment

Efficient neural architectures and hardware acceleration allow models to run on phones and embedded devices. That makes **privacy-preserving and offline ASR** possible, a major advantage for accessibility: people can use dictation, voice commands, or live captioning without necessarily streaming everything to the cloud.

2.1.5.5 Multilingual and code-switching capabilities

Large multilingual models can handle multiple languages in a single system and can better cope with code-switching. This is relevant for many users whose everyday speech mixes languages—typical in globalized and migrant contexts where access barriers are already high.

2.1.5.6 Integration into accessibility ecosystems

AI-driven ASR is now:

- Behind automatic captioning in video platforms and meeting tools.
- Integrated into live transcription and STTI workflows, where human respeakers correct and steer the output.
- Embedded in operating systems and mobile devices for dictation, voice UI, and hands-free access.

From an accessibility and translation-science point of view, AI shifts ASR from a brittle, domain-limited component to a **flexible, adaptive translator between speech and text**. That doesn't make it neutral: models still encode biases (e.g., worse performance on underrepresented accents, languages, or speech impairments), and those biases directly affect who gets accessible communication and who is left out.

For *chat-room.ch* 2.0, this means:

- The author can assume **usable ASR** for many users and treat speech-to-text as a relatively mature intersemiotic translation module, especially for standard-language, non-impaired speech.
- The author should, however, explicitly acknowledge its limitations in the concept: users with speech impairments, heavy accents, or in noisy environments may need **alternative modalities** (hums, taps, movement) that the system also translates into conversation.

2.1.6 AI-based speech recognition and its limitations for inclusive design

Recent advances in automatic speech recognition (ASR) are driven almost entirely by AI, especially deep learning. Modern systems no longer rely primarily on hand-crafted acoustic and language models, but on large neural networks trained on vast quantities of speech and text. This shift has dramatically improved recognition accuracy in everyday

conditions and enabled speech-driven interfaces to move from specialist assistive tools into mainstream devices. Dictation, voice commands, and automatic captioning are now integrated into operating systems, meeting platforms, and mobile phones.

From an inclusive design perspective, this technological leap is a double-edged sword. On the one hand, AI-based ASR makes **speech-to-text translation** far more available and responsive. Deaf and hard-of-hearing users can benefit from automatic captions in video calls and broadcasts; people with motor impairments can use dictation instead of keyboards; users with temporary constraints (hands busy, eyes occupied) gain flexible, low-friction access to digital systems. The same neural architectures can be adapted across languages and domains through fine-tuning, making it possible to support more languages and more use cases than before.

On the other hand, these systems inherit and amplify the biases and blind spots of their training data. Recognition quality remains uneven across accents, dialects, sociolects, and speaking styles. Speakers with atypical prosody or articulation—common among people with speech impairments, certain neurological conditions, or second-language users—often see lower accuracy and more unpredictable errors. Background noise and overlapping speech still degrade performance in ways that disproportionately affect those who already experience communicative barriers. In addition, the opacity of end-to-end neural models makes it difficult to diagnose or correct these biases at a fine-grained level.

There are also structural limitations that intersect with accessibility. Many high-performing ASR systems depend on cloud infrastructure, which raises concerns about privacy, connectivity, and control—especially for conversations involving sensitive health or personal information. On-device models mitigate some of these issues but are typically smaller and less capable, which can exacerbate the accuracy gap for underrepresented users. The drive to improve average performance can obscure the needs of those whose speech deviates from the statistical norm.

For an environment like *chat-room.ch* 2.0, treating AI-based ASR as a mature, universal solution would therefore be misleading. Speech recognition can serve as one powerful **intersemiotic translator**—from spoken turns into text, and into visual or sonic representations—but it must be complemented by alternative modalities. Hums, breaths, taps, device movement, or other sensor-based inputs need to be treated as first-class channels of participation rather than as fallbacks. The system's design should assume that some participants cannot or will not rely on speech, whether for technical, sensory, or personal reasons.

This has two implications for inclusive design. First, ASR should be integrated in a way that is **transparent and adjustable**: users should be able to see and correct recognition output, or to opt out of speech entirely. Second, the AI moderator and the surrounding interaction logic must not infer competence, engagement, or emotional state solely from recognized speech. Instead, they should treat speech as one modality among several, and distribute conversational agency across multiple input channels. In this way, AI-based speech recognition becomes a valuable component of a broader multimodal translation system rather than a gatekeeper of participation.

2.2 Evolution and semantic shift of “chatroom”

The author aims at a structured analysis of its origin, definition, use, and semantic evolution.

The word “**chatroom**” carries a layered technical, social, and cultural history. We associate it first with *early networked sociality*: text-based, synchronous conversation among strangers or semi-anonymous participants, mediated by a shared digital space rather than by direct personal connection.

2.2.1 Etymology and Origin

Chatroom is a compound noun:

- **chat**: informal conversation; the verb appears in English from the late 15th century, with connotations of light, spontaneous talk.
- **room**: a bounded space for gathering or activity, metaphorically extended to non-physical spaces (e.g., “newsroom,” “chat room”).

The term emerged in the **late 1980s to early 1990s**, alongside the rise of:

- Bulletin Board Systems (BBS)
- Internet Relay Chat (IRC, created in 1988)
- Early commercial online services (CompuServe, AOL, Prodigy)

Although IRC did not originally use the word “room” (it used “channels”), popular interfaces and consumer services adopted *room* as a more intuitive metaphor for non-technical users.

2.2.2 Early Definitions and Canonical Meaning

In its initial usage, a **chatroom** was defined as:

A shared online space where multiple users can exchange text messages in real time.

Key characteristics:

- **Synchronous** (live interaction)
- **Many-to-many** communication
- **Topic-based or interest-based**
- **Persistent space**, but ephemeral content
- **Pseudonymous or anonymous identity**

Importantly, chatrooms were *places*, not conversations. One *entered* and *left* them.

2.2.3 Cultural and Social Function (1990s–early 2000s)

During its peak, the chatroom functioned as:

2.2.3.1 A Social Experiment

- First large-scale experience of interacting with strangers online
- Exploration of identity, gender, age, and persona
- Early form of online role-playing and community formation

2.2.3.2 A Democratic Public Space

- Low barrier to entry
- No follower counts, no algorithmic curation
- Flat social structure

2.2.3.3 A Site of Anxiety

Simultaneously, chatrooms became associated with:

- Lack of moderation
- Online harassment

- Moral panics (especially around youth safety)
- Surveillance concerns

This duality shaped later perceptions.

2.2.4 Semantic Shift and Decline

From the mid-2000s onward, “**chatroom**” began to lose prominence.

Key reasons:

- Rise of **instant messaging** (ICQ, MSN Messenger)
- Emergence of **social networks** (Facebook, MySpace)
- Transition from *public rooms* to *private or semi-private threads*
- Multimedia communication (voice, video, images)

The conversational model shifted:

- From **spaces** → to **connections**
- From **many-to-many** → to **one-to-one or curated groups**
- From **anonymity** → to **persistent identity**

As a result, *chatroom* started to sound:

- Technically dated
- Culturally naïve
- Slightly unsafe or unregulated

2.2.5 Contemporary Usage and Reframing

Today, the word is still understood, but rarely used for mainstream platforms.

Current connotations:

- Nostalgic (early internet, Web 1.0)
- Informal or amateurish
- Sometimes suspicious (e.g., “unmoderated chatrooms”)

Functionally equivalent concepts now exist under different names:

- **Channels** (Slack, Discord)
- **Servers**
- **Groups**
- **Forums with live chat**
- **Messaging rooms**

Interestingly, AI interfaces have reintroduced the word *chat*—but not *chatroom*—with a crucial semantic difference:

- **Chat** now often implies *human–machine dialogue*, not social gathering.
- The “room” is no longer shared among peers but individualized.

2.2.6 Meaning Change in Summary

Period	Dominant Meaning
Early 1990s	Shared, public digital gathering space
Late 1990s	Social experimentation and online community
Early 2000s	Risk-laden, poorly moderated space
2010s	Obsolete or nostalgic term
2020s	Historical reference; replaced by “channels,” “groups,” “chats”

2.2.7 Conceptual Insight

The decline of *chatroom* reflects a deeper shift:

- From **place-based digital sociality** to **identity-based networks**
- From **collective presence** to **personalized communication**
- From **open commons** to **platform-governed enclosures**

In that sense, *chatroom* names not just a technology, but a **lost social model of the internet**.

2.3 Reclaiming the “chatroom” in the age of multimodal, barrier-aware AI

The historical *chatroom* emerged in the late 1980s and 1990s as a paradigmatic form of early online sociality: a bounded, shared digital space where multiple users exchanged text in real time. Its defining features were synchronicity, many-to-many communication, and a strong sense of *place*. One did not simply send messages; one *entered* and *left* rooms, often under pseudonyms, and participated in topic-oriented, semi-anonymous publics. This model offered low-threshold participation and a relatively flat social structure, but also became associated with poor moderation and various risks, contributing to its eventual decline as the dominant metaphor.

Over the ensuing decades, digital communication shifted toward identity-based networks and personalized feeds. Instant messaging, social platforms, and algorithmically curated timelines reorganized interaction around persistent profiles and one-to-one or small-group threads. In parallel, the word *chat* was repurposed for human–machine interfaces: “chat-bots” and AI assistants turned chat into a predominantly dyadic, user–system paradigm, often stripped of the spatial and collective connotations of the original chatroom. What disappeared, almost quietly, was the idea of a shared, public conversational room as a primary unit of online sociality.

During the same period, speech and accessibility technologies underwent their own transformations. Text-to-speech evolved from mechanical, rule-based voices to highly natural neural synthesis deployed in mainstream operating systems and devices. Speech recognition moved from small-vocabulary, lab-bound systems to large-vocabulary, deep-learning–based models integrated into phones, browsers, and meeting tools. Screen readers, refreshable Braille displays, subtitling for the deaf and hard of hearing, audio description for blind and low-vision audiences, and speech-to-text interpreting matured into a relatively coherent ecosystem. These tools made it increasingly feasible for people with diverse sensory and motor profiles to access and contribute to multimodal communication.

chat-room.ch 2.0 deliberately situates itself at the intersection of these two evolutions. Conceptually, it **revives the room** as a shared, bounded conversational environment: participants join tables, encounter others, and co-construct a live situation rather than interacting solely through private threads or algorithmically filtered feeds. Technically and ethically, however, it does so under contemporary conditions, treating **multimodality and accessibility as structural premises** rather than afterthoughts. The space is not limited to keyboarded text: participants can contribute through speech, typed messages, rhythmic

taps, subtle movements, or other sensor-based signals, all of which are translated into a common audiovisual fabric.

Here, the advances in speech and accessibility technologies become intersemiotic engines within the room. Speech recognition can translate spoken turns into text and visual cues; text-to-speech can render written contributions as audio for visually impaired users; minimal sensor inputs can be mapped to distinctive sonic motifs or visual traces, allowing people who cannot or prefer not to speak or type to have an audible and visible presence. Importantly, these mappings are designed with a functional and ethical focus: their purpose is to support participation, recognizability, and shared orientation, not to extract or interpret intimate data.

The AI component acts as a **multimodal moderator and interpreter** rather than as a conversational partner in the narrow chatbot sense. It monitors the dynamics of tables—detecting prolonged silence, overdominance, or fragmentation—and introduces gentle prompts, invitations, or summaries aimed at sustaining hospitable conversation. At the same time, it is explicitly constrained by barrier-aware design: it cannot assume speech is the default modality, treat ASR output as ground truth, or infer competence or engagement solely from one channel. Instead, it operates across modalities, taking text, taps, and other signals into account when assessing when and how to intervene.

In this way, *chat-room.ch 2.0* reframes the chatroom for 2026: not as a nostalgic resurrection of early Web 1.0 interfaces, nor as another individualized AI chat interface, but as a **shared, multimodal, AI-assisted space configured from the outset to accommodate different bodies and capacities**. The historical idea of the chatroom as an open, many-to-many social place is thus reactivated and critically updated through the lens of contemporary speech and accessibility technologies, with the aim of exploring how collective online presence might look if it were designed, from the ground up, as barrier-aware and sensor-rich.

2.4 Cybernetic art, sensing machines, and chat-room.ch 2.0

A third theoretical strand comes from the history of **cybernetic art** and early experiments with sensing and feedback in electronic music and performance. From the 1960s onward, artists and composers began to treat the body, sensors, and electronic systems as elements in closed feedback loops rather than as separate entities. Works using biofeedback, EEG, galvanic skin response, and environmental sensing transformed physiological processes and situational data into sound and image in real time. The human participant

became part of a sensing machine that both measured and modulated states of attention, arousal, and interaction.

In this context, Ruth Anderson's electronic and biofeedback-related work is emblematic. Her practice—alongside that of figures like Alvin Lucier, Pauline Oliveros, David Rosenboom, and others—did not use sensors simply to *record* the self but to **instantiate new experiential circuits** between body, machine, and environment. Bio-signals such as breathing, heart rate, or skin conductance were orchestrated as musical material; the performer's or audience's changing state fed back into the evolving sound field. These works framed sensing as an aesthetic and relational act: machines were not neutral instruments but active partners in shaping perception.

This lineage intersects directly with both translation theory and contemporary accessibility technologies. Cybernetic artworks can be understood as early **intersemiotic translators** of embodied states into audiovisual patterns, operating in real time and under conditions of mutual influence. Their design decisions—what to measure, how to map it to sound, how fast to respond, how much control to cede—are structurally analogous to decisions in audio description, speech-to-text interpreting, or sensor-based assistive interfaces. All of them create feedback loops in which machine readings shape human behavior, which in turn produces new readings.

chat-room.ch 2.0 can be explicitly positioned in this cybernetic art tradition, updated for the present. Like those earlier pieces, it treats sensors as mediators between embodied processes and shared audiovisual structures. Unlike many historic works, however, it frames these mediations explicitly as **conversation** and **access** rather than as one-way performance. Smartphones and other everyday devices become “soft” sensing machines: they register speech, breath, micro-movements, and simple gestures, and translate them into sonic and visual presences in the room. Participants are not passive data sources but co-creators in a dynamic feedback ecology.

The AI component plays a recognizably cybernetic role as well. It monitors global patterns (silence, domination, divergence), introduces perturbations (prompts, redistributions), and adapts mappings over time, always with the stated skopos of maintaining a hospitable conversational environment. This resonates with classic cybernetic concerns—stability, feedback, self-regulation—while embedding them in a contemporary ethics of non-dominating moderation and barrier-aware design.

Bringing this strand together with translation science and accessibility technology yields a threefold theoretical frame for the project:

- From **translation studies**, the notion of intersemiotic, multimodal, purpose-driven translation between bodies, signals, and representations.
- From **accessibility research**, the concrete techniques and norms for making multimodal content perceivable under diverse sensory and cognitive conditions.
- From **cybernetic art**, the understanding of sensing machines as aesthetic, relational, and feedback-driven systems in which humans and technologies co-constitute experience.

Within this triad, *chat-room.ch 2.0* can be read as both an homage and a critical update: it revives the shared “room” of early networked sociality and the feedback poetics of cybernetic art, but reconfigures them with contemporary AI, smartphones, and barrier-free communication as first principles rather than afterthoughts.

2.4.1 Cybernetic and Biofeedback Lineages in the Design of “chat-room.ch 2.0”

In developing, I also draw explicitly on the lineage of cybernetic and biofeedback-based artworks that treated sensing and feedback as aesthetic material. Alvin Lucier’s *Music for Solo Performer* (1965) used EEG alpha waves to excite percussion instruments, making brain activity directly audible; this inspires my decision to let low-frequency, relatively “slow” bodily signals (like breathing or periodic taps) drive sustained, textural layers in the room rather than fast, discrete events. David Rosenboom’s long-term experiments with biofeedback music, in which performers learned to modulate their physiological states to *chat-room.ch 2.0* shape sound, suggest treating sensor mappings as learnable instruments: in *chat-room.ch 2.0*, each participant’s tap or motion pattern acquires a consistent sonic identity over time, so they can develop a sense of control and authorship. Ruth Anderson’s *Centering* (1979), which routed audience galvanic skin response into electronic pitches while they watched a dancer, informs my idea of using GSR-like “arousal” proxies (e.g., density or irregularity of input) not as diagnostic measures but as compositional parameters that gently shift the intensity or brightness of a table’s sound. Taken together, these works argue for mappings that are continuous, relational, and legible to participants—principles I adopt in framing sensors in *chat-room.ch 2.0* as conversational instruments rather than opaque measurement devices.

2.5 Language, space, and interactive multimodal platforms in *chat-room.ch* 2.0

Work on “**doing space**” in interaction (Hausendorf, Jucker, Locher, Meyer, etc.) starts from the idea that space is not just a physical container in which interaction happens, but is *interactionally produced*: participants “do space” by how they position themselves, orient bodies and gaze, manage access and exclusion, and assign meaning to zones and objects. In face-to-face settings, the material environment can be heavily structured (lecture theatres, service counters), moderately structured (restaurants, staff rooms), or only weakly structured (town squares), with each degree of structuring offering different affordances and expectations for interaction (spur.uzh+1).

Jucker & colleagues extend this analysis to **interactive multimodal platforms (IMPs)**: on screens, users navigate multiple spatial levels at once—the physical screen, the layout of windows and panels, and the “virtual spaces” depicted there (rooms, channels, forums). Doing space on IMPs involves coordinating language, cursor position, visual focus, and interface elements to constitute “where we are” together: in a channel, a thread, a shared document, a breakout room, and so on. This is where Computer-Mediated Communication (CMC) pragmatics, chatspeak, and audience design intersect with spatial metaphors: users orient to “here”, “there”, “in this chat”, “in that channel” as if these were rooms, and design their language accordingly (Georgakopoulou, Herring, Beißwenger, Hausendorf) ijs+2.

chat-room.ch 2.0 explicitly leverages this line of thinking by treating the **room metaphor as an interactional architecture**, not just a nostalgic label. It is designed as a **moderately structured interaction space** in Hausendorf’s sense: more structured than a free-form town square, but less rigid than a lecture theatre. The concept of tables, the bounded number of participants per table, and the distinct audio/visual “zones” for each table are all spatial affordances: they guide how people align, to whom they feel accountable, and what they understand as “our” local conversation versus the larger event.

At the same time, as an **interactive multimodal platform**, *chat-room.ch* 2.0 adds layers of spatial complexity familiar from the IMP literature: participants inhabit a physical location (the café-like pop-up or their own home), see a **virtual layout of tables** on their screens, and are represented in **sonic and spectral space** as patterns that move and change. Doing space in this environment includes:

- Linguistic deixis (“here at Table 2”, “over there they’re very loud”).
- Multimodal deixis: sound localization (which table is sounding where in the room), visual traces tied to specific tables/participants, and turn-taking cues.
- Technological affordances: joining/leaving tables, seeing which tables are “active”, and perceiving the AI host’s interventions as coming “from the room” rather than from nowhere.

Conversation analysis and multimodal discourse analysis—Sacks, Schegloff, Goodwin, Kress & van Leeuwen—give the author tools to describe how participants actually **orient to this layered space**: how they decide when to move tables, how they treat the AI host’s prompts, how they use text vs taps vs speech as ways of claiming or yielding space. Audience design theory (Bell, Goodwin) lets the author ask how people tailor their contributions when they know they are in a semi-public table versus a more private-feeling one, or when they sense that others are listening but not yet acting.

Critically, this “space in interaction” perspective meshes with the author’s **accessibility and multimodality** frame: if space is interactionally achieved, then the question becomes *who gets to participate in doing space*. In many IMPs, that presupposes vision, rapid typing, and familiarity with dense visual layouts. *chat-room.ch 2.0* tries to lower that threshold by:

- Using simple, legible spatial structures (few tables, clear mapping to sound zones).
- Providing multiple **modes of emplacement** (speech, text, taps, simple movement), so that people with different sensory and motor profiles can still “be somewhere” in the room.
- Letting the AI host help maintain the interactional architecture (avoiding empty tables, gently redistributing attention) without centralizing control.

Integrating this literature, the author can frame *chat-room.ch 2.0* as an experiment in **how language and sensors co-constitute an interactional space** on an IMP, and how that space can be designed to be both structurally clear (spatial affordances) and accessible (multimodal, barrier-aware).

Concepts of “interactional architecture”, “emplacement”, “spatial settings on IMPs” are anchored to

- The table layout.
- The sound distribution in the physical room.
- The screen layout on the web app.

2.6 Key findings of article from Jucker et al. (2018) on IMPs for *chat-room.ch 2.0*

Jucker et al.'s article provides a precise framework for analyzing *chat-room.ch 2.0* as an **interactive multimodal platform (IMP)**, where space is not a fixed backdrop but an **interactional achievement**—"doing space"—through the interplay of material affordances, participant actions, and multimodal cues.[ppl-ai-file-upload.s3.amazonaws]

2.6.1 Core Framework: Degrees of Spatial Structuring

The authors classify physical settings by how much they pre-structure interaction:

Heavily structured (e.g., ticket counters): Rigid affordances dictate roles and turns (lining up enforces "client" position); verbal negotiation is minimal.[ppl-ai-file-upload.s3.amazonaws]

Moderately structured (e.g., museums): Furniture/chairs enable conversation but require appropriation; participants form F-formations (Kendon 1990) around exhibits.[ppl-ai-file-upload.s3.amazonaws]

Weakly structured (e.g., public squares): Minimal cues; participants must bodily negotiate co-presence or avoidance, often via unfocused interaction (Goffman 1963).[ppl-ai-file-upload.s3.amazonaws]

Application to *chat-room.ch 2.0*: The platform is a **moderately structured IMP**. Tables provide clear affordances (bounded groups of 2–4, distinct audio/visual zones), guiding participants to "enter" conversations without the rigidity of a lecture hall or the chaos of an open square. Users appropriate these by joining/leaving, much like museum-goers clustering around a showcase.

2.6.2 IMP-Specific Insights: Layered Spatial Complexity

IMPs introduce **multi-layered spaces** beyond physical settings:

Physical device/screen level.

Depicted virtual layout (e.g., rooms, chats).

Navigation challenges across layers (e.g., chat windows, pop-ups, game worlds in Twitch).[ppl-ai-file-upload.s3.amazonaws]

Examples:

Second Life: Avatars "do space" by imitating physical lecture theaters (seating in semi-circles flags "lecture" frame); deictic references ("slide on the left") require shared orientation across screen/virtual layers.[ppl-ai-file-upload.s3.amazonaws]

Twitch: Streamers navigate stacked layers (game world, platform chat, webcam overlay) via gaze shifts and cross-modal speech-to-text; viewers infer focus from verbal/nonverbal cues.[ppl-ai-file-upload.s3.amazonaws]

Application: *chat-room.ch 2.0* mirrors this with layered spaces—physical room (speakers project tables spatially), screen layout (table list → table view), and multimodal channels (text/taps/speech → shared audio/visual). Participants "do space" through:

Linguistic deixis: "Table 2 is quiet," "Who's tapping over there?"

Multimodal emplacement: Taps produce localized sounds; AI prompts anchor attention ("Table 1, anyone joining?").

2.6.3 Actionable Ties to the Design of chat-room.ch 2.0

Jucker et al. Concept	chat-room.ch 2.0 Implementation	Benefit for Accessibility
Architecture-for-interaction (pre-structured affordances)[ppl-ai-file-upload.s3.amazonaws]	Fixed 4-table layout; distinct sonic zones.	Predictable "where" reduces cognitive load; screen readers can announce "Table 2: 3 users active."
Appropriation of affordances (users adapt space)[ppl-ai-file-upload.s3.amazonaws]	Join/leave tables; choose modality (text/tap/speech).	Low-vision users hear table activity; motor-limited users tap rhythmically to claim presence.
Doing space via deixis (verbal/spatial orientation)[ppl-ai-file-upload.s3.amazonaws]	AI summaries ("Table 3 discussed X"); spectral visuals tied to tables.	Builds mutual awareness without vision; speech-to-text renders deictics audible.
Layered IMP navigation (screen/virtual shifts)[ppl-ai-file-upload.s3.amazonaws]	Simple flow: landing → table view; no pop-ups.	Minimizes disorientation; keyboard-navigable layers suit diverse devices.

2.6.4 Theoretical Positioning (2026 Update)

Despite 2018 origins, the framework endures: Recent IMPs (e.g., Discord spatial audio, VR breakout rooms) amplify layered complexity, but *chat-room.ch 2.0* innovates by embedding **barrier-aware multimodality** from the start. Unlike Twitch's chaotic layers or Second Life's avatar-heavy navigation, the piece's moderate structuring + AI moderation lowers barriers, enabling "doing space" for disabled users via alternative modalities—extending Jucker et al.'s pragmatics to inclusive design.

This positions the piece as follows:

chat-room.ch 2.0 tests "doing space" on a web-based IMP, reclaiming the chatroom as moderately structured, sensor-rich architecture where diverse bodies co-produce conversational space.

[ppl-ai-file-upload.s3.amazonaws]

2.7 Language, Space, and Interactive Multimodal Platforms in *chat-room.ch 2.0*

Jucker et al. (2018) frame space as an interactional achievement—"doing space"—where participants appropriate material affordances to constitute conversational environments, from heavily structured settings (e.g., ticket counters enforcing turn-taking via lines and barriers) to moderately structured ones (e.g., museum showcases enabling ad-hoc F-formations) and weakly structured public squares requiring bodily negotiation of co-presence. On interactive multimodal platforms (IMPs), this process gains layered complexity: users navigate physical screens, depicted virtual layouts (e.g., Second Life lecture theaters), and cross-modal cues (e.g., Twitch's stacked game/chat/webcam feeds), using deixis and meta-communication to orient "where we are together." [ppl-ai-file-upload.s3.amazonaws]

chat-room.ch 2.0 embodies a **moderately structured IMP**, akin to a museum space: its fixed four-table layout provides clear architectural affordances—bounded groups, distinct sonic/visual zones—those participants appropriate by joining/leaving and selecting modalities (text, taps, speech). Unlike heavily structured counters, tables allow flexible turn-taking without rigid queues; unlike chaotic squares, they pre-figure semi-public accountability, fostering hospitable dialogue.

This design addresses IMP-specific challenges head-on. Layered spatial navigation (physical device → screen table list → table view with spectral activity) is streamlined to avoid disorientation, while multimodal emplacement lets diverse users "do space": rhythmic taps

claim audible presence for motor-limited participants, speech-to-text renders linguistic deixis ("Table 2 is humming") accessible to deaf users, and AI prompts ("Table 1 invites responses") stabilize orientations across layers. Conversation analysis tools from the authors (Sacks, Schegloff) reveal how participants co-produce this space—e.g., responding to AI summaries or shifting tables via verbal/nonverbal cues—extending "doing space" pragmatics to barrier-aware multimodality.[ppl-ai-file-upload.s3.amazonaws]

Thus, *chat-room.ch 2.0* tests Jucker et al.'s framework in a 2026 web-based IMP: a sensor-rich, moderately structured architecture where language, sensors, and AI co-constitute inclusive conversational space, reclaiming the chatroom as an accessible social experiment.[ppl-ai-file-upload.s3.amazonaws]

3 The Author

Anet Nyffeler is an electronic media artist, applied linguist, and communication lecturer. She specialized in video art during the 1980s–2000s, exhibiting internationally in the US and Europe. Subsequently, she began a two-decade career in the film and broadcast industry, where she founded her own Ltd. company in 2001 and developed leadership and management expertise. Since 2018, she teaches organizational communication and leadership at four different institutions at the vocational college and continuing adult education levels in Switzerland. Nyffeler pursued advanced studies in communication and linguistics, publishing contributions on barrier-free communication and social media. Her practice synthesizes electronic media art, multimodal communication, and cybernetic perspectives, informing the *chat-room.ch 2.0* project and this article. She holds a BFA from Massachusetts College of Art and Design (Boston, USA) and an MA in Applied Linguistics and Organizational Communication from Zurich University of Applied Sciences.